CS 5173/6073: Deep Learning

Midterm Exam Description

The midterm exam will be conducted online using Kaggle. Please join the following link once the exam started: `https://www.kaggle.com/competitions/cs-5173-6073-deep-learning-midterm-exam`

The midterm exam will start at October 27th 12:00 pm, and all students must complete the exam individually. The deadline to submit your exam will be October 30th 11:59 pm. Fail to submit before the deadline will result 0.

Before the deadline, you must submit the following:

- **Kaggle:** submit the `submission.csv` file containing your model predictions and basic model information as specified in the Submission File section. Please read the Submission File section carefully and do not miss any file.
- **Canvas:** submit your code, model, and your model predictions.

Account registration

Before the exam start, you must register Kaggle account with your University of Cincinnati email account if you do not have one. The account username must follow the following pattern: FirstName LastName MID. For example, a student named Socrates Greek with student MID as M15832401. He will register his Kaggle account name as: Socrates Greek15832401. The proposal of this username is to identify the submission of your work on Kaggle.

Exam Task

The exam involves a supervised learning classification task. Training and testing data will be provided at the start of the exam. You are required to design, implement, and train a deep learning model independently to perform the classification within the given time. The exam should be conducted directly on the Kaggle platform. External datasets or pretrained weights are not allowed.

Problem Description

Electrocardiogram (ECG) signals are vital clinical tools for monitoring the electrical activity of the heart. Detecting abnormal heart rhythms, known as arrhythmias, plays an essential role in the diagnosis of cardiovascular disease. However, manual interpretation of ECG signals is time-consuming and requires professional expertise. The goal of this exam is to build a deep learning

model to classify heartbeats into one of the five categories. The five categories are Normal beat (N), Supraventricular ectopic beat (S), Ventricular ectopic beat (V), Fusion beat (F) and Unknown beat (Q). The objective is to predict the correct arrhythmia label for each ECG segment with minimum parameters.

Data

When you join the competition, you will find three `.csv` files.

- `train.csv` – the training set containing ECG sample and their corresponding heartbeat labels (type column). You will need to encode your labels to the following numeric values:

Label	Integer Code
F	0
N	1
Q	2
SVEB	3
VEB	4

- `test.csv` – the test set used for generating predictions. The generated prediction should follow the same numeric encoding.
- `Sample_submission.csv` – an example submission file in the correct format. Make sure to format your test.csv prediction and model parameter size formatted same as Sample_submission.csv. Fail to format will result incorrect or failed performance feedback.

Expectations

Please make sure that:

- Your code is well-structured, commented, executable, and reproducible.
- The model is efficiently designed and can be trained and tested within the time limit. The submitted results follow the required format.

Evaluation Metrics

Your performance will be evaluated based on:

- **Score**

$$\text{Score} = 0.7 \times \text{Accuracy} + 0.3 \times F_\beta - 0.02 \times \log(\text{trainable parameters}) \tag{1}$$

$$F_\beta = \frac{(1+\beta^2)(\text{Precision} \times \text{Recall})}{(\beta^2 \times \text{Precision}) + \text{Recall}}, \qquad \text{where } \beta = 2. \tag{2}$$

- **Number of trainable parameters and Estimated FLOPs**

The reference code for model statistic features is provided in the Model Statistics Function Notebook.

Scoring Policy

- A baseline performance will be provided.
- Submissions that reach or exceed the baseline will receive at least 80 points.
- Scores above the baseline will be adjusted based on ranking among all submissions.
- Submissions below the baseline will be graded proportionally using the same rule.

Academic Integrity

- Collaboration or code sharing is strictly prohibited.
- The use of AI-assisted or code-generation tools (including but not limited to ChatGPT, Claude, DeepSeek, etc.) is allowed for writing, debugging, or modifying your code or report. However, plagiarism is prohibited.
- Any violations will be handled according to the university's academic integrity policy.

Submission Requirements

All required files must be submitted before the deadline. Late submissions may not be accepted.